SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer**.** Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer**.** Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT)**.** Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

ANSWERS

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Given:

Network Topology

• PC0
• Switch (2960)
• PC1

IP Address Configuration

PC0

• IP Address = 192.168.1.1
• Subnet Mask = 255.255.255.0

PC1

• IP Address = 192.168.1.2
• Subnet Mask = 255.255.255.0

Software Used

• Cisco Packet Tracer
• Wireshark

Step 1: Design the LAN Topology

Create a simple Local Area Network using Cisco Packet Tracer.

Network Diagram

PC0 -------- Switch -------- PC1

Connect both PCs to the switch using Copper Straight-Through cables.

Answer

Network consists of

• Two Personal Computers
• One Cisco Switch
• Straight-through Ethernet cables

Step 2: Configure IP Addresses

Assign IPv4 addresses to both computers.

PC0

IP Address

192.168.1.1

Subnet Mask

255.255.255.0

PC1

IP Address

192.168.1.2

Subnet Mask

255.255.255.0

Since both hosts belong to the same subnet, they can communicate directly through the switch.

Answer

PC0 = 192.168.1.1

PC1 = 192.168.1.2

Subnet Mask = 255.255.255.0

Step 3: Verify Connectivity Using Ping

Open Command Prompt on PC0.

Execute the command

ping 192.168.1.2

Expected Output

Reply from 192.168.1.2

Packets Sent = 4

Packets Received = 4

Packets Lost = 0

This confirms successful communication between the two hosts.

Answer

• Ping Successful
• Connectivity Verified

Step 4: Capture Packets Using Wireshark

Start packet capture in Wireshark.

Perform the ping operation again.

Stop the capture after receiving the replies.

Locate the Ethernet II frame generated during ICMP communication.

Answer

Wireshark successfully captures the Ethernet frames exchanged between PC0 and PC1.

Step 5: Identify Ethernet Frame Fields

From the captured Ethernet frame, identify the following fields.

Source MAC Address

The physical address of the sender (PC0).

Example

00:0C:85:12:34:56

Destination MAC Address

The physical address of the receiving device (PC1).

Example

00:0C:85:65:43:21

EtherType

Indicates the protocol encapsulated inside the Ethernet frame.

For IPv4

EtherType = 0x0800

Frame Check Sequence (FCS)

Used for error detection.

The receiving device recalculates the FCS to verify that the frame was transmitted without errors.

Answer

- Source MAC Address = MAC address of PC0
- Destination MAC Address = MAC address of PC1
- EtherType = 0x0800 (IPv4)
- Frame Check Sequence (FCS) = Used for error detection

Step 6: Explain the Role of the Data Link Layer

The Data Link Layer (OSI Layer 2) is responsible for reliable communication between devices within the same Local Area Network.

Functions of the Data Link Layer

- Uses MAC addresses to uniquely identify network devices.
- Encapsulates IP packets into Ethernet frames.
- Detects transmission errors using the Frame Check Sequence (FCS).
- Ensures that frames are delivered only to the intended destination device.
- Works together with switches to forward frames efficiently.

Switch Operation

The switch learns the MAC addresses of connected devices and stores them in its MAC Address Table.

Whenever a frame arrives, the switch forwards it only to the port associated with the destination MAC address, reducing unnecessary network traffic.

Answer

The Data Link Layer ensures successful frame delivery by using MAC addresses for local communication and FCS for error detection.

Final Answers

Network Topology

• PC0 — Switch — PC1

IP Configuration

• PC0 = 192.168.1.1 /24
• PC1 = 192.168.1.2 /24

Connectivity Test

• Ping Successful
• 4 Packets Sent
• 4 Packets Received
• 0 Packet Loss

Ethernet Frame Fields

• Source MAC Address = MAC Address of PC0
• Destination MAC Address = MAC Address of PC1
• EtherType = 0x0800 (IPv4)
• Frame Check Sequence (FCS) = Error Detection Field

Functions of the Data Link Layer

• Uses MAC addressing for local communication.
• Encapsulates packets into Ethernet frames.
• Detects transmission errors using FCS.
• Enables switches to forward frames correctly.
• Ensures reliable frame delivery within the LAN.

Explanation

A simple LAN was created using two computers connected through a switch in Cisco Packet Tracer. Both hosts were configured with IPv4 addresses belonging to the same subnet and connectivity was verified successfully using the ping command. During communication, Wireshark captured the Ethernet frames, allowing the identification of important fields such as the Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence. The Data Link layer plays a crucial role by assigning MAC addresses to devices, encapsulating network-layer packets into Ethernet frames, detecting transmission errors using FCS, and enabling switches to forward frames only to the intended destination.

Conclusion

The experiment successfully demonstrated the working of a simple Local Area Network and verified communication between two hosts using ICMP ping. Wireshark analysis confirmed the structure of Ethernet frames and the importance of MAC addressing. The Data Link layer ensures accurate and reliable frame delivery within the local network through MAC addressing, frame encapsulation, switching, and error detection using the Frame Check Sequence.

ANSWERS

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Given:

Network Topology

LAN 1

- PC0
- Switch0

LAN 2

- PC1
- Switch1

Interconnecting Device

• Router

Software Used

• Cisco Packet Tracer
• Wireshark

IP Address Configuration

PC0

• IP Address = 192.168.1.2
• Subnet Mask = 255.255.255.0
• Default Gateway = 192.168.1.1

Router Interface G0/0

• 192.168.1.1

Router Interface G0/1

• 192.168.2.1

PC1

• IP Address = 192.168.2.2
• Subnet Mask = 255.255.255.0
• Default Gateway = 192.168.2.1

Step 1: Design the Network

Construct two LANs connected through a router.

Network Diagram

PC0 — Switch0 — Router — Switch1 — PC1

Answer

The network contains two separate LANs connected using a router.

Step 2: Configure IPv4 Addressing

Assign the IP addresses to all devices.

Answer

PC0 = 192.168.1.2

Router G0/0 = 192.168.1.1

Router G0/1 = 192.168.2.1

PC1 = 192.168.2.2

Subnet Mask = 255.255.255.0

Step 3: Verify End-to-End Connectivity

Execute

ping 192.168.2.2

Expected Result

Packets Sent = 4

Packets Received = 4

Packet Loss = 0%

Answer

Communication between the two LANs is successful.

Step 4: Capture ICMP Packets

Start Wireshark capture.

Execute ping again.

Stop packet capture after communication completes.

Answer

Wireshark captures ICMP Echo Request and Echo Reply packets.

Step 5: Analyze IPv4 Header Fields

Source IP Address

192.168.1.2

Destination IP Address

192.168.2.2

Time-To-Live (TTL)

Typically 128 (Windows) or 64 (Linux)

Protocol Number

1 (ICMP)

Header Length

20 Bytes

Answer

- Source IP = 192.168.1.2
- Destination IP = 192.168.2.2
- TTL = 128 (Typical)
- Protocol = 1 (ICMP)
- Header Length = 20 Bytes

Step 6: Explain the Role of the Network Layer

The Network Layer (OSI Layer 3) performs the following functions:

- Assigns logical IP addresses.
- Determines the best routing path.
- Forwards packets between different networks.
- Performs packet forwarding using routers.
- Supports communication across multiple LANs.

Final Answers

Network Topology

- Two LANs connected using a router.

IPv4 Configuration

- PC0 = 192.168.1.2
- Router G0/0 = 192.168.1.1
- Router G0/1 = 192.168.2.1

• PC1 = 192.168.2.2

IPv4 Header Fields

• Source IP Address = 192.168.1.2
• Destination IP Address = 192.168.2.2
• TTL = 128
• Protocol Number = 1 (ICMP)
• Header Length = 20 Bytes

Explanation

The router connects two different LANs and forwards packets between them using IPv4 addresses. Wireshark captures ICMP packets generated by the ping command and displays important IPv4 header fields such as Source IP Address, Destination IP Address, TTL, Protocol Number, and Header Length. The Network layer is responsible for logical addressing, routing, and packet forwarding between different networks.

Conclusion

The experiment successfully demonstrated communication between two different LANs through a router. IPv4 addressing and routing enabled successful packet forwarding, while Wireshark verified the Network layer header information during ICMP communication

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Given:

Network Topology

• PC0
• PC1
• PC2
• Cisco Switch

Software Used

• Cisco Packet Tracer

• Wireshark

Step 1: Construct the LAN

Connect multiple PCs to a switch using Copper Straight-Through cables.

Answer

A Local Area Network containing multiple hosts is successfully created.

Step 2: Configure IP Addresses

PC0

192.168.1.1

PC1

192.168.1.2

PC2

192.168.1.3

Subnet Mask

255.255.255.0

Answer

All devices belong to the same subnet.

Step 3: Clear the ARP Cache

Open Command Prompt.

Execute

arp -d

This removes previously stored MAC-IP mappings.

Answer

ARP cache cleared successfully.

Step 4: Generate ARP Traffic

Execute

ping 192.168.1.2

The sender first broadcasts an ARP Request before sending ICMP packets.

Answer

ARP Request and ARP Reply packets are generated.

Step 5: Capture Packets Using Wireshark

Start Wireshark.

Perform ping.

Stop packet capture.

Locate the ARP packets.

Answer

Wireshark displays ARP Request followed by ARP Reply.

Step 6: Analyze ARP Fields

ARP Request

Sender IP

192.168.1.1

Sender MAC

MAC Address of PC0

Target IP

192.168.1.2

Target MAC

00:00:00:00:00:00 (Unknown)

ARP Reply

Sender IP

192.168.1.2

Sender MAC

MAC Address of PC1

Target IP

192.168.1.1

Target MAC

MAC Address of PC0

Answer

ARP Request asks for the MAC address corresponding to the destination IP, and ARP Reply provides the required MAC address.

Step 7: Explain Address Resolution Protocol (ARP)

ARP performs the following functions:

- Maps IPv4 addresses to MAC addresses.
- Broadcasts ARP Requests.
- Receives ARP Replies.
- Stores mappings in the ARP Cache.
- Enables successful Ethernet communication.

Final Answers

Topology

- Multiple hosts connected through a switch.

ARP Request

- Sender IP = 192.168.1.1
- Target IP = 192.168.1.2

ARP Reply

• Sender IP = 192.168.1.2
• Target IP = 192.168.1.1

Functions of ARP

• Resolves IP addresses into MAC addresses.
• Updates ARP Cache.
• Enables successful frame transmission.

Explanation

Before transmitting data over Ethernet, a sender must know the MAC address of the destination device. ARP resolves this by broadcasting an ARP Request containing the target IP address. The destination device replies with its MAC address through an ARP Reply. The sender stores this mapping in the ARP cache and uses it for future communication. Wireshark captures both ARP Request and Reply packets, allowing analysis of sender and target MAC/IP addresses.

Conclusion

The experiment demonstrated how ARP resolves logical IPv4 addresses into physical MAC addresses before communication begins. Wireshark successfully captured ARP Request and Reply packets, proving that ARP is an essential protocol for communication within a Local Area Network.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Given:

Network Topology

• PC0
• Router
• PC1

Software Used

• Cisco Packet Tracer
• Wireshark

IP Address Configuration

PC0

• IP Address = 192.168.1.2
• Subnet Mask = 255.255.255.0
• Default Gateway = 192.168.1.1

Router Interface G0/0

• 192.168.1.1

Router Interface G0/1

• 192.168.2.1

PC1

• IP Address = 192.168.2.2
• Subnet Mask = 255.255.255.0
• Default Gateway = 192.168.2.1

Step 1: Design the Network

Create a network in Cisco Packet Tracer.

Network Diagram

PC0 -------- Router -------- PC1

Answer

The network consists of two computers connected through a router.

Step 2: Configure IP Addresses

Assign the required IPv4 addresses to both PCs and router interfaces.

Answer

PC0 = 192.168.1.2

Router G0/0 = 192.168.1.1

Router G0/1 = 192.168.2.1

PC1 = 192.168.2.2

Step 3: Verify Connectivity

Open Command Prompt on PC0.

Execute

ping 192.168.2.2

Expected Output

Packets Sent = 4

Packets Received = 4

Packet Loss = 0%

Answer

Successful communication between both hosts confirms proper routing.

Step 4: Capture ICMP Traffic

Start Wireshark.

Perform the ping operation.

Stop the capture.

Locate ICMP packets.

Answer

Wireshark captures ICMP Echo Request and Echo Reply packets.

Step 5: Analyze ICMP Fields

Echo Request

Type = 8

Code = 0

Identifier = Unique Ping Identifier

Sequence Number = 1,2,3,4

Echo Reply

Type = 0

Code = 0

Identifier = Same as Request

Sequence Number = Same as Request

Answer

ICMP Request

• Type = 8
• Code = 0

ICMP Reply

• Type = 0
• Code = 0

Identifier

• Used to match Request and Reply packets.

Sequence Number

• Identifies each ping packet.

Step 6: Calculate Round Trip Time (RTT)

RTT represents the total time taken for a packet to travel from the sender to the receiver and back.

Example

Reply from 192.168.2.2

Time = 2 ms

Therefore,

Round Trip Time

RTT = 2 ms

Answer

RTT = 2 ms (Example observed during ping)

Step 7: Explain the Role of ICMP

ICMP performs the following functions:

- Tests network connectivity.
- Reports transmission errors.
- Measures response time.
- Supports Ping and Traceroute.
- Helps diagnose routing and communication problems.

Final Answers

Network Topology

- PC0 — Router — PC1

Connectivity

- Ping Successful

ICMP Echo Request

- Type = 8
- Code = 0

ICMP Echo Reply

- Type = 0
- Code = 0

Identifier

- Matches request and reply packets.

Sequence Number

- Identifies each ping packet.

Round Trip Time

• RTT = 2 ms

Functions of ICMP

• Connectivity verification
• Error reporting
• Network troubleshooting
• Performance measurement

Explanation

A simple routed network was created using two computers connected through a router. After configuring IPv4 addresses and default gateways, connectivity was verified using the ping command. Wireshark captured ICMP Echo Request and Echo Reply packets, allowing analysis of important ICMP header fields such as Type, Code, Identifier, and Sequence Number. The Round Trip Time (RTT) indicates the time required for a packet to travel to the destination and return. ICMP plays a significant role in network diagnostics by identifying communication problems and verifying connectivity.

Conclusion

The experiment successfully demonstrated ICMP communication between two hosts connected through a router. Wireshark confirmed the exchange of Echo Request and Echo Reply packets, while RTT measurement verified network performance. ICMP is an essential protocol for connectivity testing and troubleshooting in IP networks.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Given:

Network Topology

• Client PC
• Switch
• Web Server

Software Used

• Cisco Packet Tracer
• Wireshark

IP Address Configuration

Client PC

• IP Address = 192.168.1.2
• Subnet Mask = 255.255.255.0

Web Server

• IP Address = 192.168.1.10
• Subnet Mask = 255.255.255.0

HTTP Service

• Enabled on the Web Server

Step 1: Create the Client-Server Network

Connect the Client PC and Web Server through a switch.

Network Diagram

Client PC -------- Switch -------- Web Server

Answer

A client-server network is successfully created.

Step 2: Configure IP Addresses

Assign IPv4 addresses to both devices.

Answer

Client PC = 192.168.1.2

Web Server = 192.168.1.10

Subnet Mask = 255.255.255.0

Step 3: Enable HTTP Service

Open the Server configuration.

Enable HTTP service.

Save the configuration.

Answer

HTTP service is successfully enabled on the server.

Step 4: Access the Web Page

Open the web browser on the client.

Enter

http://192.168.1.10

The webpage loads successfully.

Answer

The client successfully accesses the hosted webpage.

Step 5: Capture Packets Using Wireshark

Start packet capture.

Access the webpage again.

Stop the capture.

Answer

Wireshark captures TCP and HTTP packets.

Step 6: Identify TCP Three-Way Handshake

Connection Establishment

Step 1

Client → Server

SYN

Step 2

Server → Client

SYN + ACK

Step 3

Client → Server

ACK

Answer

TCP Three-Way Handshake

- SYN
- SYN-ACK
- ACK

Step 7: Identify HTTP Messages

HTTP GET Request

Sent by the client requesting the webpage.

HTTP Response

Sent by the server containing the webpage.

Answer

Application Layer Messages

- HTTP GET Request
- HTTP Response (200 OK)

Step 8: Explain Reliable TCP Delivery

TCP ensures reliable communication by:

- Establishing a connection before transmission.
- Acknowledging received packets.
- Retransmitting lost packets.
- Delivering packets in the correct order.

• Performing error checking.

Final Answers

Network

• Client PC — Switch — Web Server

HTTP Service

• Enabled Successfully

TCP Three-Way Handshake

• SYN
• SYN-ACK
• ACK

HTTP Communication

• HTTP GET Request
• HTTP Response (200 OK)

Functions of TCP

• Connection establishment
• Reliable delivery
• Acknowledgments
• Ordered data transfer
• Error recovery

Explanation

A client-server network was created with a web server providing HTTP services. The client accessed the hosted webpage using a web browser. Wireshark captured the TCP three-way handshake followed by the HTTP GET request and the HTTP response. TCP establishes a reliable connection before data transfer, acknowledges received packets, retransmits lost packets, and ensures that data arrives in the correct order. This guarantees reliable delivery of HTTP traffic between the client and server.

Conclusion

The experiment successfully demonstrated HTTP communication over TCP. The captured packets confirmed the TCP three-way handshake, HTTP GET request, and HTTP response. TCP provides reliable communication through connection establishment, acknowledgments, retransmissions, and ordered delivery, ensuring accurate and efficient web communication.